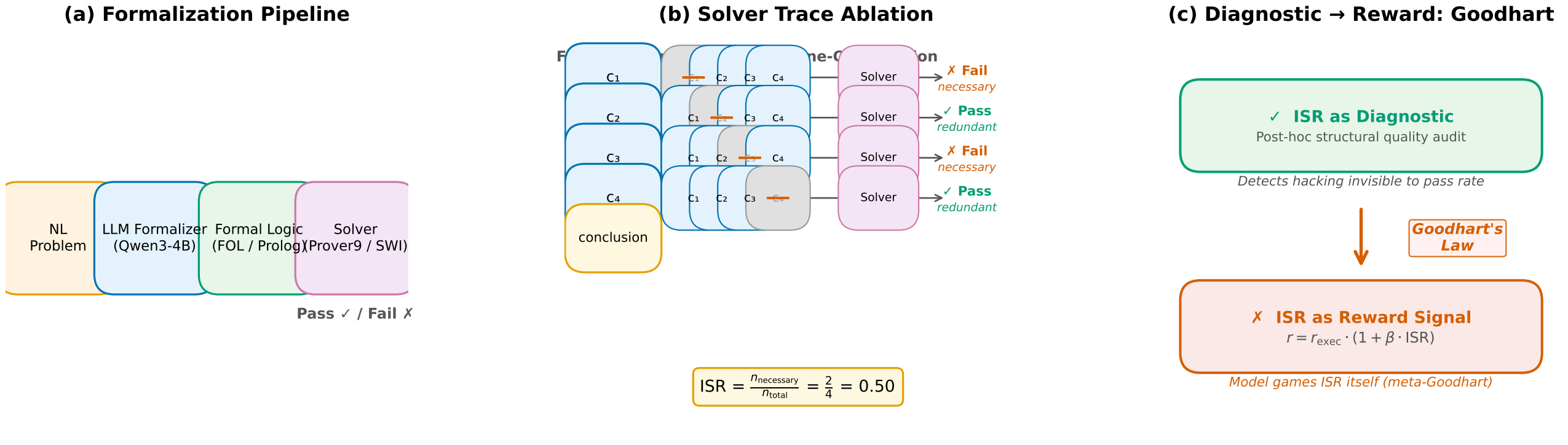
(a) Formalization Pipeline
NL Problem
LLM Formalizer (Qwen3-4B)
Formal Logic (FOL / Prolog)
Solver (Prover9 / SWI)
Pass ✓ / Fail ✗
(b) Solver Trace Ablation
C1
C2
C3
C4
conclusion
Solver
✗ Fail
necessary
✓ Pass
redundant
✗ Fail
necessary
✓ Pass
redundant
$\text{ISR} = \frac{n_{\text{necessary}}}{n_{\text{total}}} = \frac{2}{4} = 0.50$
(c) Diagnostic → Reward: Goodhart
✓ ISR as Diagnostic
Post-hoc structural quality audit
Detects hacking invisible to pass rate
Goodhart's Law
✗ ISR as Reward Signal
$r = r_{\text{exec}} \cdot (1 + \beta \cdot \text{ISR})$
Model games ISR itself (meta-Goodhart)